

Agricultural Performance and Seasonal Analysis Using Data Analytics

A Data Analytics Project Report

Dataset: Crop Yield in Indian States (Kaggle)

Coverage: 1997–2020 | 55 crops | 30 states | 6 seasons

Records analyzed: 19,689 rows x 10 columns

Report type: Business/Decision-Oriented Data Analytics Report

Abstract / Executive Summary

This report presents a business-oriented data analytics study of Indian agricultural performance using a state-wise crop dataset covering 1997–2020 (19,689 records across 55 crops, 30 states, and 6 cropping seasons). The analysis follows a Raw Data → Information → Insights → Decision → Action framework: after data-quality checks and cleaning, we compute headline KPIs, analyze production and yield trends over time, benchmark crop and state performance, examine seasonal cropping patterns, and statistically test the relationship between rainfall, fertilizer, pesticide usage, and yield.

Total reported production grew by approximately 200.2% between 1997 and 2019 while cultivated area grew only 1.8%, indicating that growth has been driven primarily by productivity gains rather than land expansion. Correlation analysis shows that rainfall, fertilizer, and pesticide usage have negligible linear correlation with yield at the pooled, all-crop level (all $|r| < 0.03$ for rainfall and per-hectare inputs), a finding that argues for crop-specific rather than blanket agricultural interventions. The report concludes with concrete risks, opportunities, and recommended actions for agricultural planners and agribusiness stakeholders.

Problem Statement

Agricultural stakeholders — government planners, agribusinesses, and researchers — require an evidence-based understanding of how crop production and yield have evolved across crops, states, and seasons in India, and whether widely assumed yield drivers (rainfall, fertilizer, pesticide) show a measurable statistical relationship with actual outcomes. Without this analysis, resource allocation and policy decisions risk being based on assumption rather than data.

Objectives

- Identify overall national production and yield trends.
- Evaluate crop-wise performance in terms of production volume and yield.
- Evaluate state/regional performance.
- Analyze seasonal (cropping-season) patterns in production and yield.
- Examine the statistical relationship between rainfall, fertilizer, pesticide usage and yield.
- Identify risks, opportunities and actionable recommendations for agricultural decision-making.

Dataset Description and Source

The dataset used is the publicly available Crop Yield in Indian States Dataset from Kaggle (kaggle.com/datasets/akshatgupta7/crop-yield-in-indian-states-dataset), provided as `crop_yield.csv`. It contains state-wise, crop-wise, season-wise agricultural records for India covering the years 1997 to 2020.

Property	Value
Rows	19,689
Columns	10
Years covered	1997–2020
Number of crops	55
Number of states	30
Number of seasons	6
Missing values	0
Duplicate rows	0
Records in 2020 (partial year)	37

Data Dictionary

Column	Description
Crop	Name of the crop
Crop_Year	Year the crop was harvested
Season	Cropping season (Kharif, Rabi, Whole Year, Autumn, Summer, Winter)
State	Indian state
Area	Cultivated area (hectares)
Production	Total production (unit varies by crop; most in tonnes, coconut in nut-count)
Annual_Rainfall	Annual rainfall recorded for that state (mm)
Fertilizer	Total fertilizer used (kg)
Pesticide	Total pesticide used (kg)
Yield	Production per unit area (Production / Area)

Data Cleaning & Preparation

The raw dataset contained no missing values and no duplicate rows. However, categorical columns (Crop, Season, State) contained trailing whitespace (e.g. 'Kharif ') which was stripped prior to analysis to prevent duplicate category groupings.

Zero-Production Records — Investigated, Not Deleted

112 records (0.569% of the data) report zero production despite non-zero cultivated area. Rather than removing these as errors, they were investigated: they are concentrated in specific state/crop combinations (top states: Haryana, Chhattisgarh, Kerala; top crops: Jute, Sannhamp, Horse-gram), consistent with localized harvest failures (drought, flood, or pest damage) rather than data-entry errors. These rows are retained for area/production and risk analysis, and excluded only from yield-based averages (since a Yield of 0 in a failed-harvest record is not informative about typical attainable yield).

Extreme Production Values — Explained, Not Removed

The largest production values in the dataset all belong to Coconut, exceeding six billion units in some state-years. Investigation shows this is not a data error: coconut production in this dataset is recorded as a nut count, while most other crops are recorded in tonnes/quintals. This unit difference legitimately produces values orders of magnitude larger than other crops, and coconut is treated distinctly (and excluded from cross-crop production/yield comparisons on a shared scale) rather than being deleted as an outlier.

Partial-Year Handling (2020)

The dataset contains only 37 records for 2020, versus roughly 900–1,100 records per year in most other years. 2020 is therefore excluded from year-over-year trend charts and growth-rate calculations throughout this report, and is never presented as a complete annual data point.

Methodology

- Data loaded with pandas; data types, missing values and duplicates checked first.
- Categorical whitespace stripped; zero-production and extreme-value records investigated and documented rather than blindly dropped.
- A 'yield-safe' subset (Production > 0) used specifically for yield-based averages and correlations.
- KPIs computed on the full cleaned dataset (or yield-safe subset for the yield KPI).
- Crop- and state-level rankings computed with a minimum record-count threshold (≥ 20 records) to avoid small-sample noise in 'top yield' claims.
- Correlation analysis (Pearson r) used to describe, not to imply causal, relationships between environmental/input variables and yield; fertilizer and pesticide converted to per-hectare intensity to avoid conflating scale-of-farm effects with productivity effects.

- All charts generated directly from the cleaned dataset using matplotlib/seaborn.

KPI Analysis

KPI	Value
Total Production (tonnes/units)	323,607,247,726.00
Average Yield	80.41
Total Cultivated Area	3,542,574,242.80
Average Annual Rainfall (mm)	1,437.76
Total Fertilizer Usage	474,570,118,804.54
Total Pesticide Usage	961,775,229.94
Number of Crops	55
Number of States	30

These KPIs give the executive-level baseline for the rest of the analysis: national scale of production, land use, and input consumption across all crops, states and years in scope.

Yearly Trend Analysis

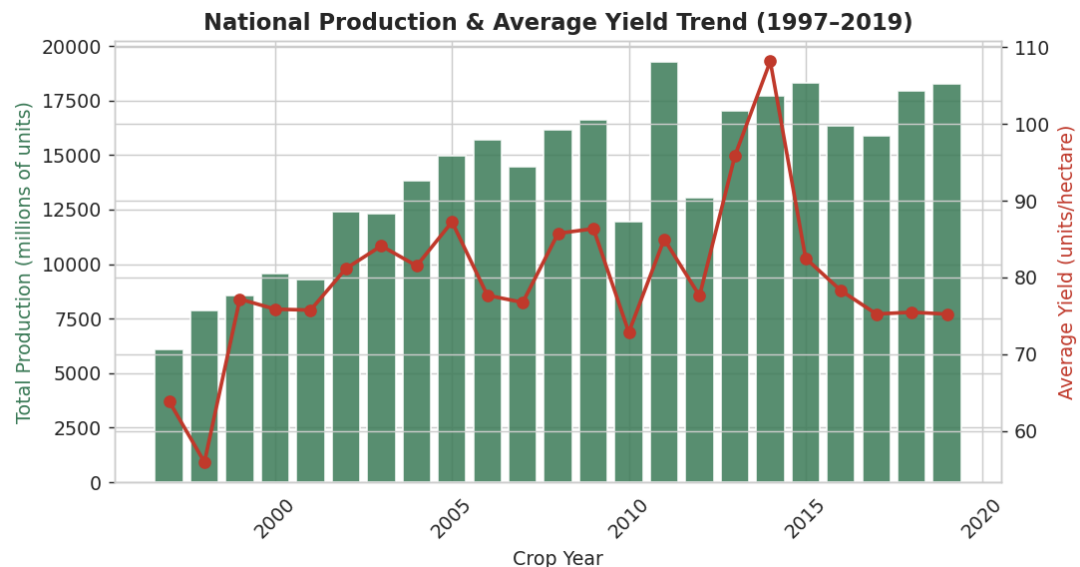


Figure 1. National total production (bars, left axis) and average yield (line, right axis) by crop year, 1997-2019. 2020 excluded (partial year).

Total reported production increased by approximately 200.2% from 1997 to 2019, while total cultivated area grew by only 1.8% over the same period. This gap indicates that production growth has been driven predominantly by yield/productivity improvements and shifts in crop mix, rather than by expanding the land under cultivation. Average yield shows substantial year-to-year volatility, consistent with weather-driven variation rather than a smooth upward trend.

Crop Analysis

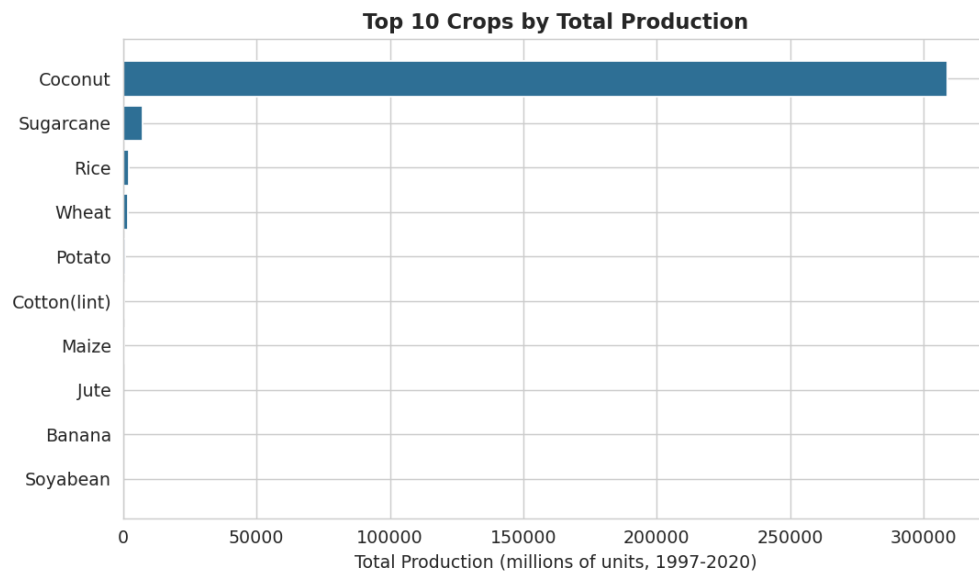


Figure 2. Top 10 crops by total reported production, 1997-2020.

Coconut dominates total production by a wide margin. As explained in the Data Cleaning section, this reflects coconut being measured in nut-count rather than tonnes, not a claim that coconut is the most economically significant crop. Rice, wheat and sugarcane — India's major staple and cash crops — are the next largest contributors by reported volume.

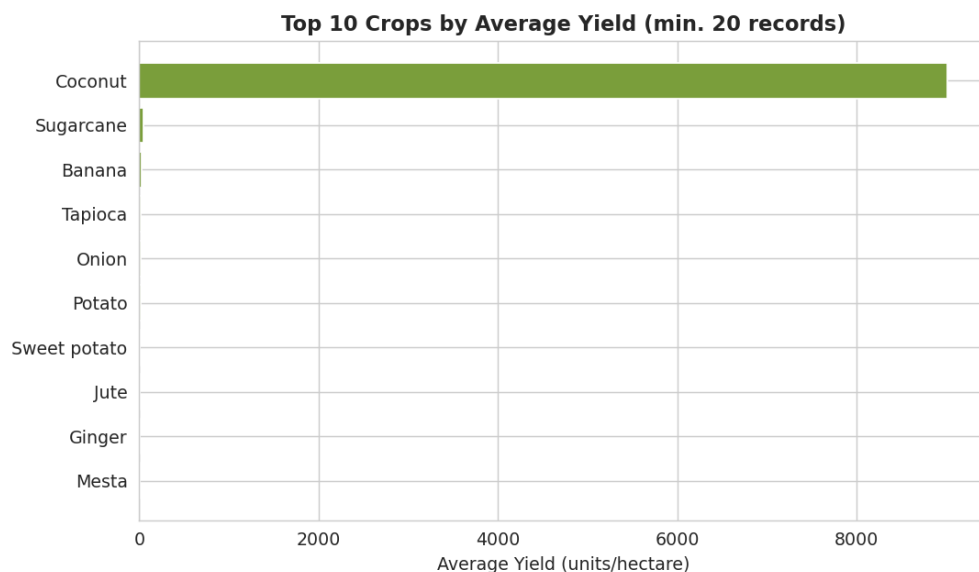


Figure 3. Top 10 crops by average yield among crops with at least 20 records (unreliable small-sample crops excluded).

Excluding the coconut unit-scale effect, Sugarcane, Banana and Tapioca show the strongest average yields among reliably sampled crops. At the opposite end, pulse and oilseed crops such as Cardamom, Niger seed, Moth, and Horse-gram show the lowest average yields — expected agronomically, since these are typically rainfed, low-yield-per-hectare crop types rather than a sign of poor cultivation.

State / Regional Analysis

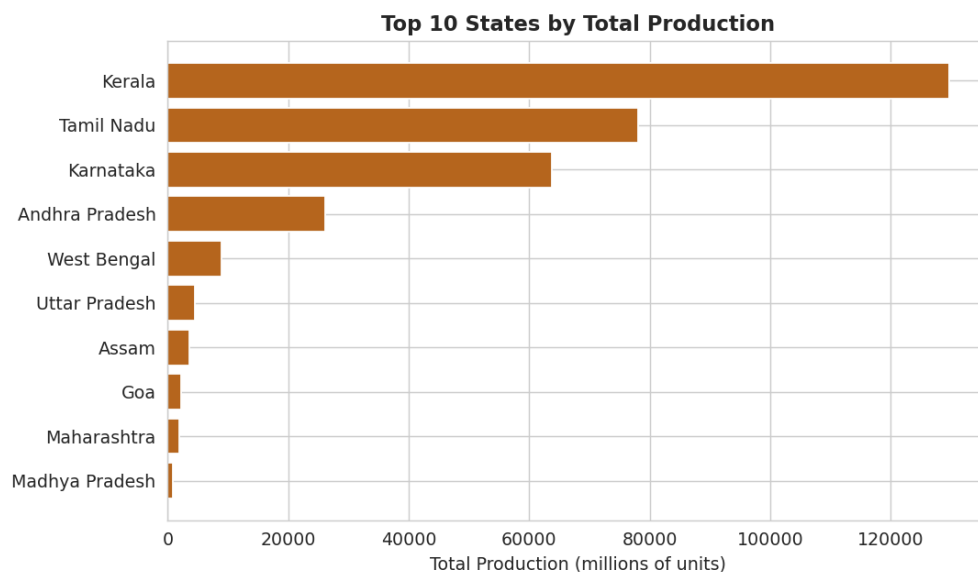


Figure 4. Top 10 states by total reported production, 1997-2020.

Kerala, Tamil Nadu and Karnataka lead national production totals, again strongly influenced by large coconut output in these states. Uttar Pradesh, West Bengal and Andhra Pradesh remain major contributors driven by rice, wheat and sugarcane. At the other end, hilly/rainfed states such as Sikkim, Jammu & Kashmir, Himachal Pradesh, Chhattisgarh and Odisha show the lowest average yields among reliably sampled states — consistent with terrain, lower irrigation penetration, and a crop mix weighted toward low-yield pulses and millets.

Seasonal Analysis

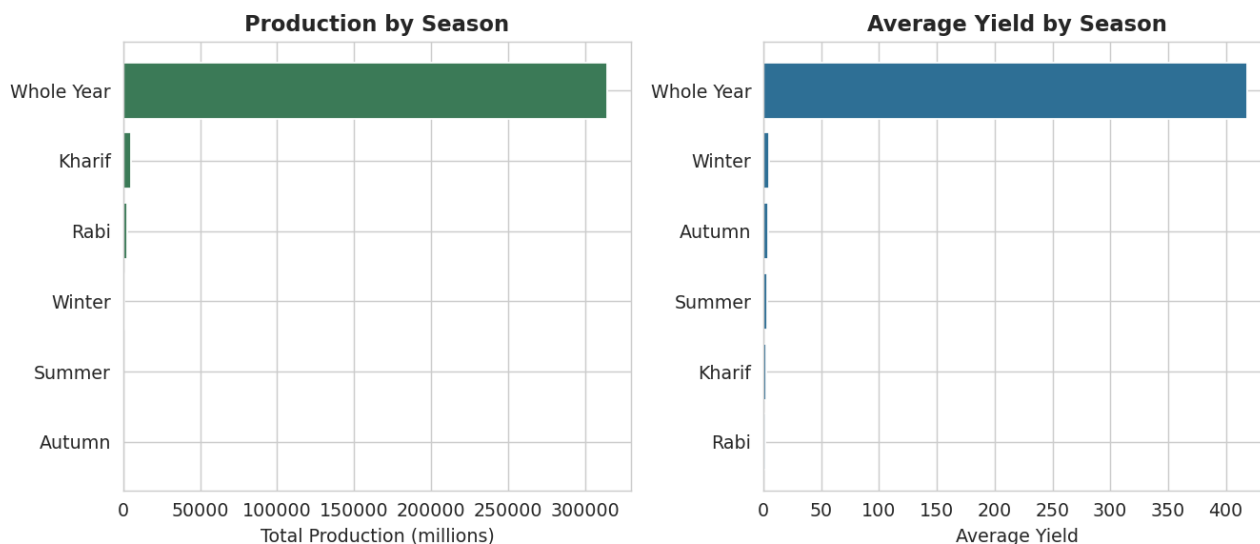


Figure 5. Total production (left) and average yield (right) by cropping season.

Kharif (the monsoon-sown season) accounts for the largest cultivated area of any season in this dataset, reflecting India's monsoon-dependent cropping calendar, followed by Rabi (the winter season). The Whole Year category — covering perennial crops such as coconut, sugarcane and banana that are harvested continuously rather than in one narrow window — shows by far the highest average yield and a large share of total production. Autumn, Summer and Winter are comparatively minor cropping windows by both area and output in this dataset.

Environmental / Input Analysis (Rainfall, Fertilizer, Pesticide vs Yield)

Correlation, not causation: the statistics below describe linear association only. A weak correlation does not mean a variable is unimportant to farming outcomes — it can also mean the effect is masked when pooling many different crops and regions together, or that the relationship is non-linear or conditional on other factors (soil, irrigation, variety, management).

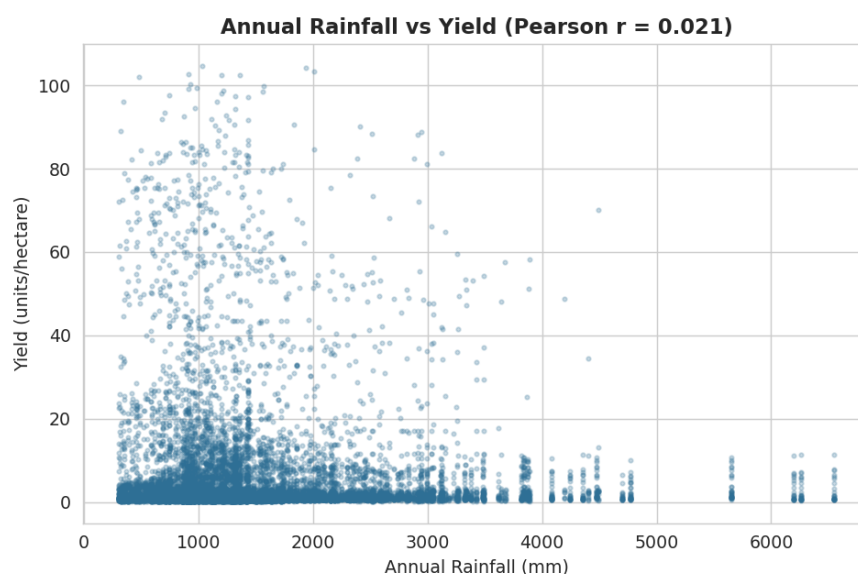


Figure 6. Annual rainfall vs. yield, all crops pooled (Pearson $r = 0.021$).

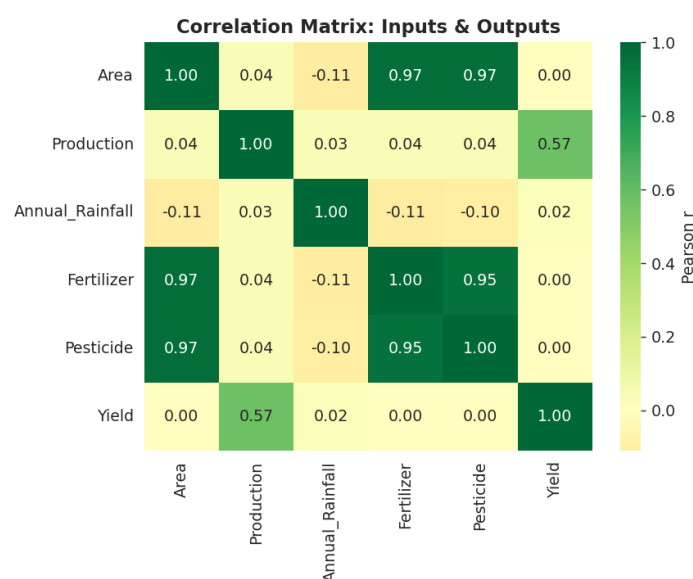


Figure 7. Correlation matrix across Area, Production, Annual Rainfall, Fertilizer, Pesticide and Yield.

At the pooled, all-crop level, Annual Rainfall ($r = 0.021$), Fertilizer-per-hectare ($r = 0.002$) and Pesticide-per-hectare ($r = -0.002$) all show near-zero linear correlation with yield. Notably, the correlation matrix shows that Area, Fertilizer and Pesticide totals are very strongly correlated with each other ($r \approx 0.95$ – 0.97) — this is expected, since a larger cultivated area mechanically requires more total fertilizer and pesticide, and is precisely why per-hectare intensity (not raw totals) was used when relating inputs to yield. The practical implication is that a single national-level relationship between input use and yield cannot be responsibly claimed from this pooled analysis; crop-specific and region-specific studies are needed to isolate genuine input-yield relationships.

Important Findings / Insights

- Production grew ~200.2% (1997→2019) while cultivated area grew only ~1.8%: growth is productivity-led, not land-expansion-led.

- Coconut dominates raw production/yield totals due to its distinct nut-count unit, not necessarily genuine cross-crop superiority.
- Sugarcane, Banana and Tapioca are the strongest reliable yield performers (excluding the coconut unit effect).
- Kharif is the largest cropping season by area; Whole Year (perennial) crops show the highest yield.
- Rainfall, fertilizer-per-hectare and pesticide-per-hectare show negligible pooled correlation with yield — this argues for crop-specific analysis, not a conclusion that these inputs don't matter.
- 112 records (0.569%) show zero production despite cultivated area — legitimate harvest-failure signals concentrated in specific state/crop combinations.

Risks

- Rainfall dependence in the Kharif season exposes the largest share of cultivated area to monsoon-driven weather risk.
- Recurring zero-production incidents (112 records) concentrated in specific states (e.g. Haryana, Chhattisgarh, Kerala) and crops (e.g. Jute, Sannhamp, Horse-gram) indicate localized harvest-failure risk worth monitoring.
- States with the highest rainfall variability (coefficient of variation) — Gujarat, Meghalaya, Manipur, Haryana, Tamil Nadu — face elevated year-to-year production risk.
- Structurally low-yield states in hilly/rainfed terrain (Sikkim, J&K, Himachal Pradesh, Chhattisgarh, Odisha) are unlikely to improve from uniform, one-size-fits-all national policy.

Opportunities

- Continued yield-improving investment (irrigation, seed variety, soil management) is a more promising growth lever than land expansion, given historical trends.
- High-yield, reliably performing crops (sugarcane, banana, tapioca) could be promoted in agro-climatically suitable low-yield states currently dominated by low-yield pulses.
- Because pooled input-yield correlation is negligible, there is room to audit fertilizer/pesticide input efficiency crop-by-crop, potentially reducing input costs without a yield penalty.

Actionable Recommendations

- Prioritize irrigation and water-management investment in Kharif-heavy, high rainfall-variability states (e.g. Gujarat, Haryana, Tamil Nadu) to reduce weather-driven production risk.
- Commission crop-specific (not pooled) regression/agronomic studies on fertilizer and pesticide efficiency before making blanket input-usage recommendations.
- Investigate the recurring zero-production state-crop combinations for root cause (pest, drought, market withdrawal) and build early-warning support for those regions.
- Report coconut and other count-based-unit crops separately in any cross-crop KPI or dashboard to avoid misleading headline production/yield figures.
- Exclude or clearly flag 2020 in any year-over-year reporting, since it is a partial year in this dataset.

Limitations

- This is a descriptive/correlational analysis of secondary, state-reported data; it does not control for soil quality, irrigation access, farm size, market prices, or crop variety.
- 2020 contains only 37 records and is excluded from annual trend and growth-rate calculations; it should not be treated as a complete year elsewhere either.
- Correlation coefficients reported here describe linear association only and must not be interpreted as causal effects.
- Pooling all crops together in a single correlation can mask genuine crop-specific relationships between inputs and yield; a follow-up crop-specific or regional regression study is recommended.

- Production units differ by crop (notably coconut in nut-count vs. tonnes for most crops), which limits direct cross-crop comparison of raw production/yield figures.

Conclusion

Between 1997 and 2019, India's reported agricultural production grew by approximately 200.2% while cultivated area grew by only 1.8%, indicating that growth in this dataset has been led by productivity gains rather than land expansion. Coconut dominates raw production and yield totals largely due to its distinct measurement unit, not genuine cross-crop superiority. Kharif is the dominant cropping season by area, while perennial (Whole Year) crops show the highest yields. At the pooled level, rainfall, fertilizer and pesticide show negligible linear correlation with yield, a finding that calls for crop- and region-specific analysis rather than uniform policy. The state, crop, and risk findings in this report point toward targeted opportunities — irrigation investment in high-variability states, input-efficiency audits, and monitoring of recurring harvest-failure patterns — as the most promising, evidence-based next steps for agricultural decision-makers.

Technologies Used

- Python 3
- pandas, numpy — data manipulation and cleaning
- matplotlib, seaborn — data visualization
- scipy — statistical calculations (Pearson correlation)
- Jupyter Notebook — reproducible analysis environment
- ReportLab — PDF report generation

Project Structure / Submission Files

File	Description
Agricultural_Performance_Analysis.ipynb	Complete, executed Python analysis notebook
crop_yield.csv	Source dataset (place alongside the notebook)
requirements.txt	Python dependencies with compatible version ranges
Agricultural_Performance_Analysis_Report.pdf	This detailed project report
README.md	Project overview, setup and run instructions